

Math Education With Large Language Models: Supplementary Materials

Additional Figures

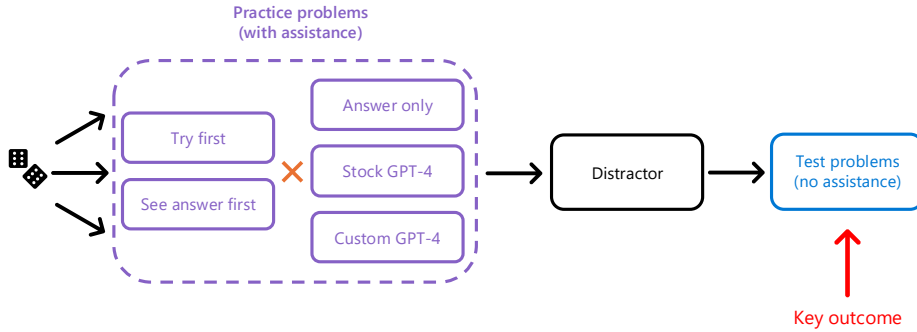


Figure 1: Schematic of the Experiment 1 design.

Qualitative analysis of strategies

After solving the test problems we asked participants to describe the strategy they used for each problem in their own words. We did this to learn more about whether they adopted the strategies they were shown in the practice rounds, and if so how effective different strategies were in improving performance. To investigate this we used an iterative coding scheme to categorize each of the free text responses. First, we manually labeled a sample of 200 responses to determine the set of strategies that participants tried for each question type and whether they explicitly mentioned following the strategy provided in the practice round or not. For instance, for the average speed calculation problem shown in Figure 1, strategy labels included the *Stock LLM*'s "harmonic mean" approach, the *Customized LLM*'s "make up a distance" strategy, and some alternative strategies that were not directly shown to participants (e.g., "algebra with $D=RT$ ", which can be time consuming but yields a correct response, and "averaging speeds", which results in an incorrect answer). We also included labels for when participants indicated that they had simply guessed. Then we

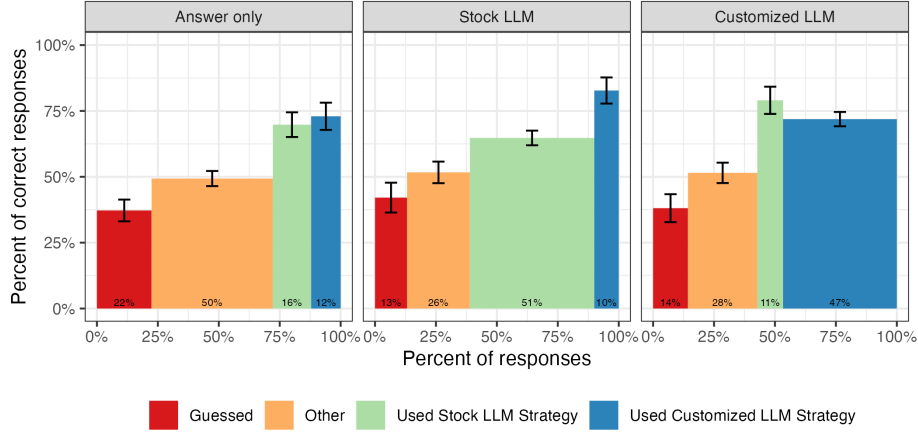


Figure 2: Strategies used by participants for solving test problems. The width of each rectangle (annotated on the bottom of each rectangle) shows the percent of responses that used each strategy in each condition, while the height shows the corresponding average accuracy when that strategy was used. Error bars show one standard error above and below these averages. Increased performance is largely due to participants adopting the strategies they were exposed to by the LLM.

created a few shot prompt for GPT-4 with examples of each strategy label for each question type and ran each response through the GPT-4 API to generate labels at scale for all 2400 responses (2 responses for each of the 1200 participants). To check the quality of these labels, we verified that the GPT-generated labels agreed with the manual labels on the sample of 200 responses that we coded ourselves.

With strategy labels generated for each question type, we analyzed which strategies participants adopted and the efficacy of these strategies, aggregated across questions, as shown in Figure 2. Specifically, we looked at whether participants followed one of the strategies recommended by the LLMs, if they used an alternative strategy, or if they simply guessed.

LLM system prompts

You are a tutor designed to help people understand and perform better on the types of problems you are given.

When solving problems where there are unknown numbers, try making up a values for those number to simplify the math. Choose these numbers so that any subsequent arithmetic works out nicely (e.g., by choosing numbers that are whole number multiples of all numbers mentioned in the problem to avoid decimal points or fractions in subsequent divisions).

Here are some strategies to use when arriving at and explaining answers, if applicable:

- When solving problems that involve speeds, emphasize the " $D = RT$ " strategy for "Distance = Rate * Time"
- When solving problems that involve averages (other than speeds), emphasize the " $T = AN$ " strategy for "Total = Average * Number"
- When given a problem with two unknown numbers involving their sum and difference, suggest the strategy of starting with half their sum and distributing the difference

Explain how to solve the given problem in a way which makes it easiest for the learner to understand and remember so that they are able to apply it to a similar problem in future. Avoid solutions that require memorization of concepts and avoid complex notation. The learner should be able to develop some intuition to solve similar problems.

In the end, summarize the solution in minimal number of lines so that the learner is able to remember the method in future.

Figure 3: The system prompt for the Customized LLM in Experiment 1 (not visible to participants).

System prompt:

You are a math tutor who provides concise explanations. Here is an example input and output.

Input:

$5x / (x + 24) = 3x / 3$. Find the value of x (that is not 0).

Output:

To find the value of x that is not 0, we can solve the equation by cross-multiplication. Given the equation:

$$3x/(x+24) = 5x/5$$

First, cancel common values (i.e., the 5s on the right and the x 's on both sides, noting that x cannot equal 0):

$$3/(x+24) = 1$$

Now, cross-multiply and solve for x :

$$3 = x + 24$$

$$x = 3 - 24 = -21$$

Query:

$7x/(x - 64) = 4x/4$. Find the value of x (that is not 0).

Output:

To find the value of x that is not 0, we can solve the equation by cross-multiplication. Given the equation:

$$7x/(x - 64) = 4x/4$$

First, cancel common values (i.e., the 4s on the right and the x 's on both sides, noting that x cannot equal 0):

$$7/(x - 64) = 1$$

Now, cross-multiply and solve for x :

$$7 = x - 64$$

$$x = 7 + 64$$

$$x = 71$$

So the value of x that is not 0 is 71.

Figure 4: System prompt and output for the Customized LLM (not visible to participants) *without Error* condition in Experiment 2.